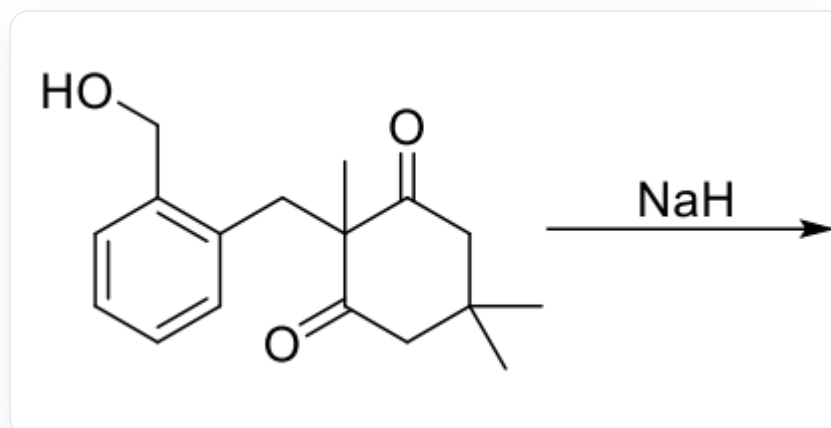


Question



The figure shows an organic reaction, the reactant is OCC1=C(CC2(C)C(CC(C)(C)CC2=O)=O)C=CC=C1, the condition is adding NaH, the product is unknown

For the reaction shown in the figure, which of the following statements is correct:

- A.** The first step of this reaction generates an enolate intermediate.
- B.** The first step of this reaction is the nucleophilic attack of a hydride ion on the carbonyl group.
- C.** The product has three six-membered rings.
- D.** The reaction process involves a six-membered ring carbanion intermediate.
- E.** The product is a sodium salt.
- F.** The product contains a six-membered ring lactone.
- G.** The product contains an eight-membered ring ether.

H. The product can react with metallic sodium.

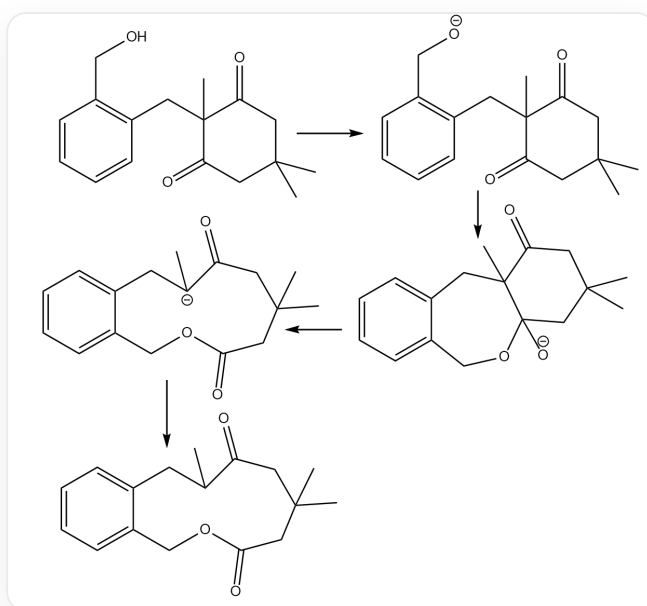
I. None of the above options are correct.

Answer

Correct Answer: **I**

Detailed Explanation

The reaction intermediates and products are shown in the figure:



Reaction intermediates and products

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Sodium hydride is a strong base but has weak nucleophilicity, so the first step reacts with the most active proton, namely the hydroxyl hydrogen, to generate the oxyanion $[O^-]CC1=CC=CC=C1CC2(C)C(CC(C)(C)CC2=O)=O$, options A and B are incorrect.

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CHECKPOINT

1 PTS

Generation of oxyanion [O-]CC1=CC=CC=C1CC2(C)C(CC(C)(C)CC2=O)=O

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Subsequently, the oxyanion attacks the carbonyl group, generating the tetrahedral intermediate O=C1[C@]2(C)CC3=CC=CC=C3CO[C@@]([O-])2CC(C)(C)C1.

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CHECKPOINT

1 PTS

Generation of tetrahedral intermediate O=C1[C@]2(C)CC3=CC=CC=C3CO[C@@]([O-])2CC(C)(C)C1

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The next reaction of the tetrahedral intermediate is carbon-carbon bond cleavage. The cleaved carbon-carbon bond should generate the most stable carbanion. Option D is correct. The carbon-carbon bond connecting the carbonyl group and the quaternary carbon atom cleaves, and the generated carbanion is relatively stable due to the electron-withdrawing effect of another adjacent carbonyl group. The carbanion is O=C(CC(C)(C)C1)[C-](C)CC2=CC=CC=C2COC1=O.

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CHECKPOINT

1 PTS

Generation of carbanion O=C(CC(C)(C)C1)[C-](C)CC2=CC=CC=C2COC1=O

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In the workup process, the carbanion abstracts a proton to obtain the product. The product structure is O=C(CC(C)(C)C1)C(C)CC2=CC=CC=C2COC1=O, which is an electrically neutral product with only one six-membered ring,

a 11-membered ring lactone and an ether bond, so all options are incorrect.

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CHECKPOINT

1 PTS

Product structure is O=C(CC(C)(C)C1)C(C)CC2=CC=CC=C2COC1=O

CHECKPOINT

1 PTS

The product molecule has an 11-membered ring structure, which contains an ester group and an ether bond